

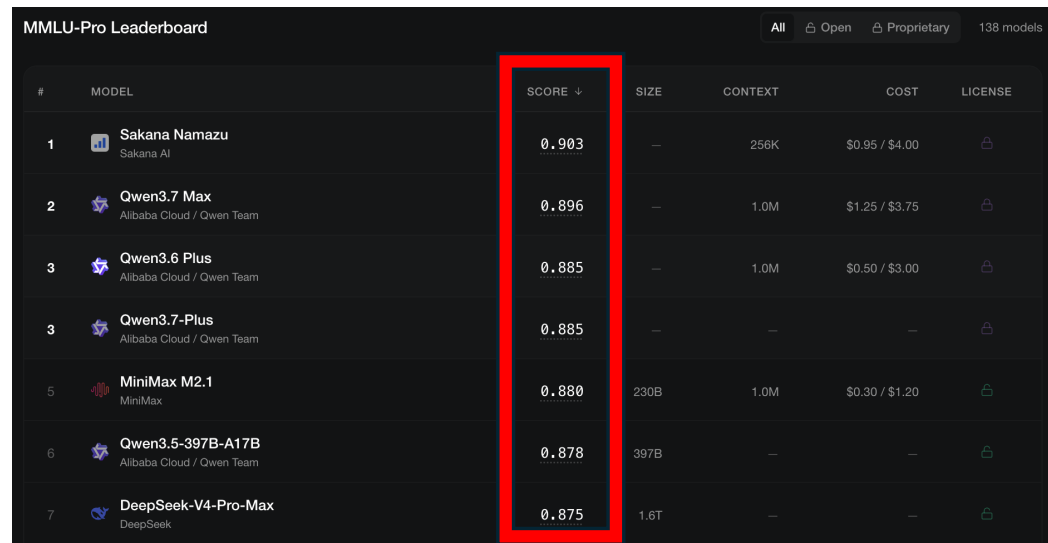
Uncertainty-Aware Bayesian Prompt Imputation and Recovery for Sparse Multi-Prompt LLM Evaluation

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LLM Evaluation

LLM benchmarks are usually reported as a single accuracy number under one prompt.

e.g.



MMLU-Pro Leaderboard

All Open Proprietary 138 models

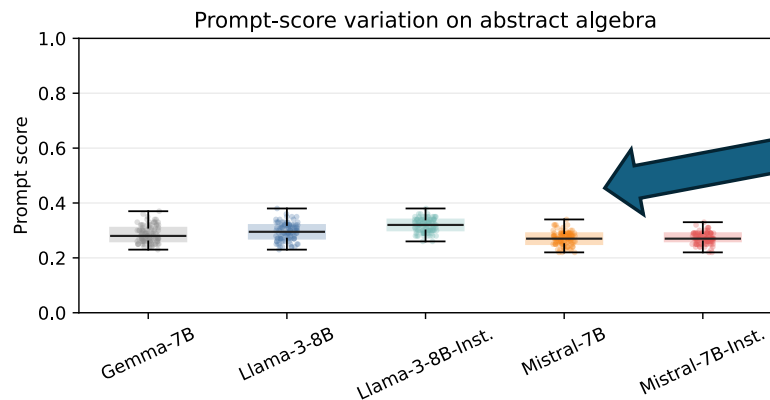
#	MODEL	SCORE ↓	SIZE	CONTEXT	COST	LICENSE
1	 Sakana Namazu Sakana AI	0.903	—	256K	\$0.95 / \$4.00	
2	 Qwen3.7 Max Alibaba Cloud / Qwen Team	0.896	—	1.0M	\$1.25 / \$3.75	
3	 Qwen3.6 Plus Alibaba Cloud / Qwen Team	0.885	—	1.0M	\$0.50 / \$3.00	
3	 Qwen3.7-Plus Alibaba Cloud / Qwen Team	0.885	—	—	—	
5	 MiniMax M2.1 MiniMax	0.880	230B	1.0M	\$0.30 / \$1.20	
6	 Qwen3.5-397B-A17B Alibaba Cloud / Qwen Team	0.878	397B	—	—	
7	 DeepSeek-V4-Pro-Max DeepSeek	0.875	1.6T	—	—	

Reference: <https://llm-stats.com/benchmarks/mmlu-pro>

Motivation

If you evaluate the same with two slightly different prompt wordings, you can get very different accuracy scores.

e.g.



e.g. in MMLU abstract algebra, the prompt scores have an average range of 0.128.

Prompt Variation does matter in LLM Evaluation.

So multi-prompt evaluation might be needed.

Diagram 1: MMLU (abstract algebra) prompt score over 100 prompts across five LLM Model

Problem

If we want to evaluate “I prompts over J items”,
we need run $I \times J$ times of the model call.

This exhaustive evaluation is very expensive,
particularly if we want to evaluate the model on high level subjects.

Is it possible for us to evaluate only a sample part of the exhaustive evaluation,
while we can accurately estimate the full multi-prompt performance?

Research Question

1. Given only a small observed subset of prompt-item outcomes, can we recover the full distribution of prompt scores?
2. How confident should we be on the recovery?

Related Work

1. Prompt Sensitivity is a real problem:

Mizrahi et al. (2024) show that evaluating with a single prompt template can give misleading results. Prompt sensitivity should be considered.

2. PromptEval:

It is the closest method to us. It models prompt-item correctness using a Rasch item-response model and uses it to recover the prompt scores from a sparse subset of evaluation.

Mizrahi, M., Kaplan, G., Malkin, D., Dror, R., Shahaf, D., & Stanovsky, G. (2024). *State of what art? A call for multi-prompt LLM evaluation*. arXiv:2401.00595.

Maia Polo, F., Xu, R., Weber, L., Silva, M., Bhardwaj, O., Choshen, L., Melo de Oliveira, A. F., Sun, Y., & Yurochkin, M. (2024). *Efficient multi-prompt evaluation of LLMs*. NeurIPS.

Data

- MMLU correctness results released by PromptEval (Maia Polo et al. 2024)
- 5 models (Gemma-7B, Llama-3-8B, Llama-3-8B-Instruct, Mistral-7B-v0.1, Mistral-7B-Instruct-v0.2), 10 MMLU subjects, and 100 prompt templates per subject.

We try to align our experiment with PromptEval

Reference: <https://github.com/felipemaiapolo/prompteval>

Item-Prompt Evaluation

Imagine a matrix: 100 prompt templates (rows) x benchmark items (columns).

Each cell:

$$Y_{ij}^* \in \{0,1\} \quad \text{did prompt } i \text{ get item } j \text{ correct?}$$

Realised Prompt Score (Accuracy of one prompt across all items):

$$S_i^* = \frac{1}{J} \sum_{j=1}^J Y_{ij}^*$$

The distribution of scores across all 100 prompts:

$$F^*(x) = \frac{1}{I} \sum_{i=1}^I \mathbf{1}\{S_i^* \leq x\}$$

Item-Prompt Evaluation

For example:

Say we have 5 prompts, with scores:

$$S^* = \{0.40, 0.55, 0.60, 0.75, 0.90\}$$

Then,

$$F^*(0.60) = \frac{4}{5} = 0.8$$

Because 4 out of 5 scored less than 60.

This shows the full spread of prompt scores, so it gives more information.

Set up

We want to estimate the full $I \times J$ matrix by only observing a subset of cells $\mathcal{E}$, with the budget B .

$$\mathcal{E} \subseteq \mathcal{I} \times \mathcal{J}, \quad |\mathcal{E}| = B$$

The observed outcomes are $Y_{\mathcal{E}} = \{Y_{ij}^* : (i, j) \in \mathcal{E}\}$

In our experiment, we have 100 prompts, estimating with $B \in \{200, 400, 800, 1600\}$ observed outcomes.

BayPIRE: Bayesian Rasch Model

Posterior draw
IRT model

We add a hierarchical Bayesian prior on the additive Rasch model.

$$Y_{ij} \mid \theta \sim \text{Bernoulli}(p_{ij}), \quad \text{logit}(p_{ij}) = \alpha + a_i - b_j,$$

where α : global performance level, a_i : prompt effect, b_j : item difficulty.

$$\theta = (\alpha, a^{\text{raw}}, b^{\text{raw}}, \sigma_a, \sigma_b).$$

PE-Rasch	BayPIRE
Maximum likelihood	Hierarchical Bayesian priors
One fitted parameter vector	Full posterior distribution
No shrinkage prior	Regularisation / shrinkage
Point estimates	Point estimates + uncertainty

Point Recovery:

For each posterior draws $\theta^{(1)}, \dots, \theta^{(M)}$,
$$p_{ij}^{(m)} = \text{logit}^{-1} \left(\alpha^{(m)} + a_i^{(m)} - b_j^{(m)} \right).$$

Averaging the posterior probability,

$$\hat{p}_{ij} = \frac{1}{M} \sum_{m=1}^M p_{ij}^{(m)}.$$

Then reconstruct

$$\hat{Y}_{ij} = \begin{cases} Y_{ij}^*, & (i, j) \in \mathcal{E}, \\ \hat{p}_{ij}, & (i, j) \in \mathcal{E}^c. \end{cases}$$

Our prompt score is

$$\hat{S}_i = \frac{1}{J} \sum_{j=1}^J \hat{Y}_{ij}.$$

Posterior predictive unc:

For each posterior draw:

$$\tilde{Y}_{ij}^{(m)} = \begin{cases} Y_{ij}^*, & (i, j) \in \mathcal{E}, \\ \text{Bernoulli} \left(p_{ij}^{(m)} \right), & (i, j) \in \mathcal{E}^c. \end{cases}$$

and

$$S_i^{(m)} = \frac{1}{J} \sum_{j=1}^J \tilde{Y}_{ij}^{(m)}.$$

Then,

$$\left[Q_{0.025} \left(\{S_i^{(m)}\}_{m=1}^M \right), Q_{0.975} \left(\{S_i^{(m)}\}_{m=1}^M \right) \right].$$

Evaluation Metrics (1)

Let S_i^* be the realised prompt score and $\hat{S}_i$ the recovered prompt score.

Definition 4.1 (Prompt-level point recovery). Mean absolute error and root mean squared error are

$$\text{MAE} = \frac{1}{I} \sum_{i=1}^I |\hat{S}_i - S_i^*|, \quad \text{RMSE} = \left\{ \frac{1}{I} \sum_{i=1}^I (\hat{S}_i - S_i^*)^2 \right\}^{1/2}.$$

Measures how far our recovered prompt score is from the true score.

Definition 4.2 (Distributional recovery). Distributional recovery is evaluated using

$$W_1(\hat{F}, F^*) = \int_0^1 |\hat{Q}(u) - Q^*(u)| du, \quad \text{1st Wasserstein distance}$$

where $\hat{Q}$ and Q^* are the corresponding quantile functions.

Measures how different our recovered distribution of scores is from the true one.

Evaluation Metrics (2)

Definition 4.3 (Hidden-cell probability prediction). On $\mathcal{E}^c$, the hidden-cell log loss is

$$\text{LogLoss} = -\frac{1}{|\mathcal{E}^c|} \sum_{(i,j) \in \mathcal{E}^c} [Y_{ij}^* \log \hat{p}_{ij} + (1 - Y_{ij}^*) \log(1 - \hat{p}_{ij})],$$

and the hidden-cell brier score is

$$\text{Brier} = \frac{1}{|\mathcal{E}^c|} \sum_{(i,j) \in \mathcal{E}^c} (Y_{ij}^* - \hat{p}_{ij})^2.$$

Measures how accurate our predicted probabilities are on missing cells.

Definition 4.4 (Posterior predictive uncertainty). For BayPIRE, let $[L_i^{0.95}, U_i^{0.95}]$ be the nominal 95% posterior predictive interval for prompt i . Predictive coverage and mean interval width are

$$\text{Coverage}_{0.95} = \frac{1}{I} \sum_{i=1}^I \mathbf{1}\{S_i^* \in [L_i^{0.95}, U_i^{0.95}]\}, \quad \text{Width}_{0.95} = \frac{1}{I} \sum_{i=1}^I (U_i^{0.95} - L_i^{0.95}).$$

Check whether our 95% intervals contain the truth for 95% of time, and whether the width increase as more cells are observed

Results: point & distributional recovery

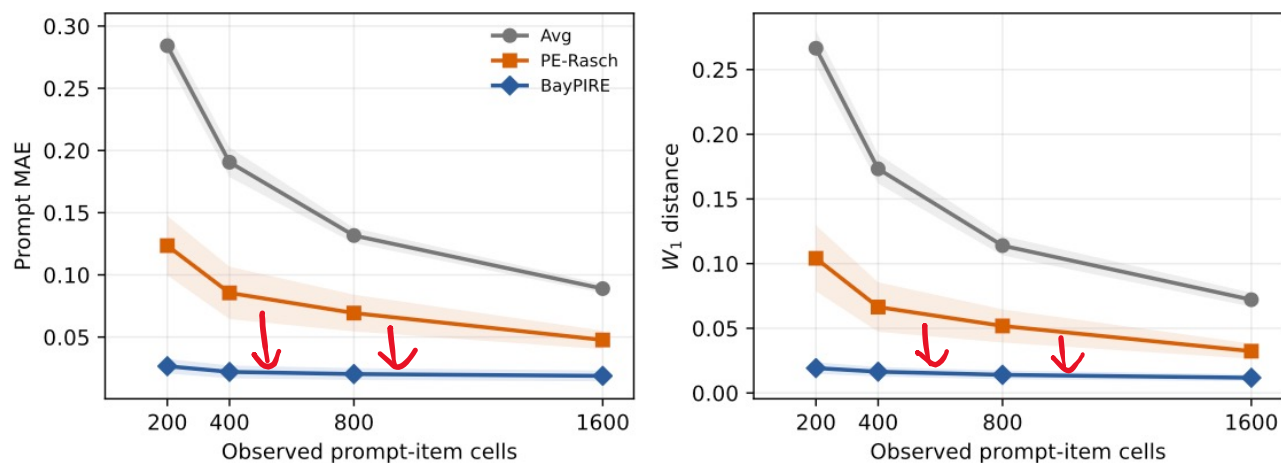


Figure 1 Prompt-level and distributional recovery under two-way balanced masking.

BayPIRE cuts prompt-score error over than half compared to PromptEval's method, even at the smallest budget!!!

Results: hidden-cell prediction

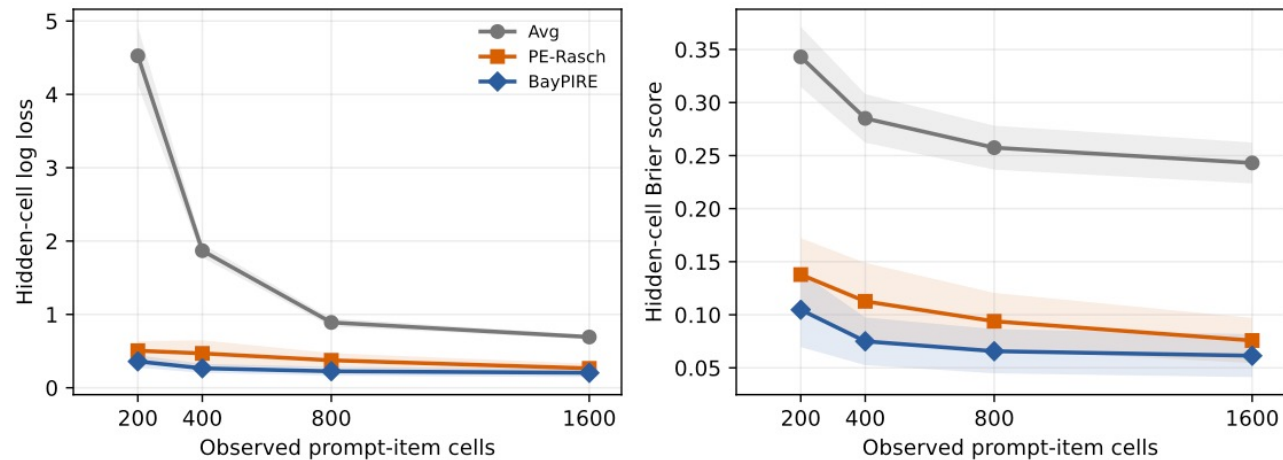


Figure 2 Hidden-cell probabilistic prediction under balanced masking.

BayPIRE also gives **better fitting probability estimates** on unobserved outcomes!!

Results: uncertainty quantification

Our 95% intervals cover the true score about 98-100% of the time.
Maybe a bit conservative, but it is not over confidence.

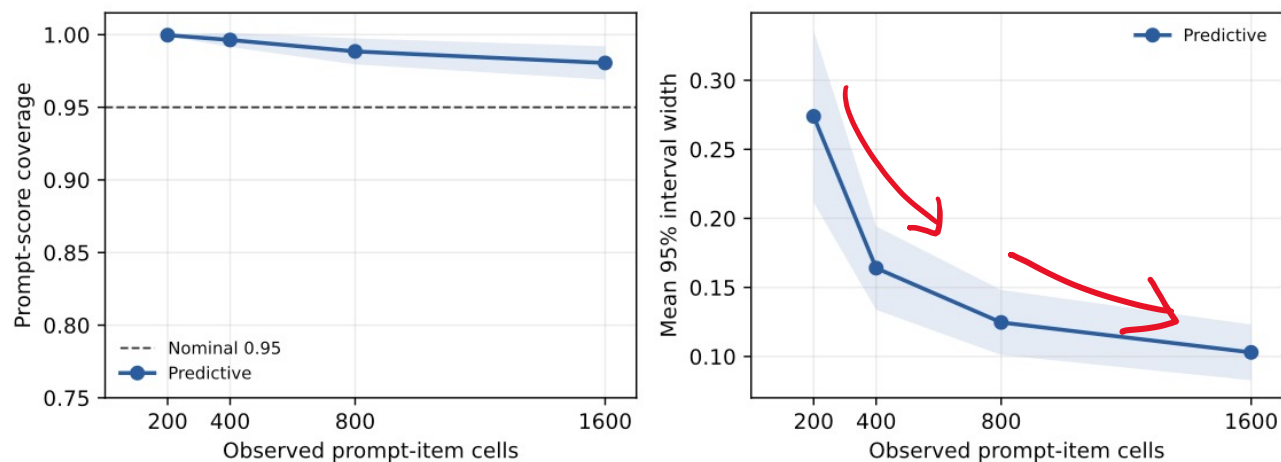


Figure 3 BayPIRE posterior predictive coverage and interval width under balanced masking.

Interval width decreases monotonically as budget increases.

Conclusion

With so few observed cells per prompt, MLE estimates can be noisy or unstable. BayPIRE's priors let each prompt and item borrow information from the rest of the grid. This can shrink noisy estimates toward a more stable value. That's why the gap between the two methods is largest at the smallest budget, and shrinks as more data comes in.

BayPIRE extends sparse multi-prompt recovery from a point-estimate problem to an uncertainty-aware one.

It also outperforms the existing baselines on every metric we tested.

Limitations:

1. Doesn't consider model randomness (Temperature >0).
2. Intervals are too conservative.
3. MCMC is more expensive than point estimators.

Future:

1. Consider the hierarchical interaction between prompt and item.
2. Model repeated stochastic generation (non-deterministic).